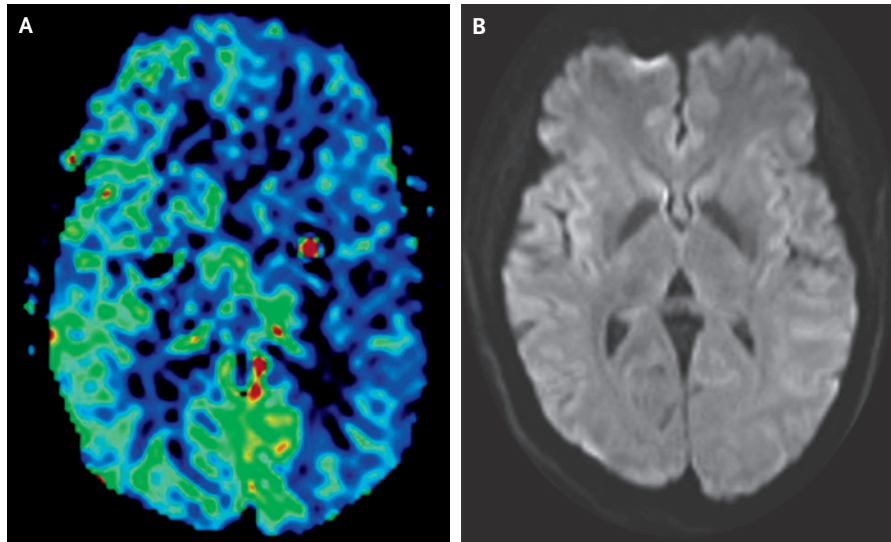


IMAGES IN CLINICAL MEDICINE

Hemiplegic Migraine



A 46-YEAR-OLD WOMAN WITH A HISTORY OF MIGRAINE WITH VISUAL AURA presented to the emergency department with a 3-hour history of a throbbing, diffuse headache. At the onset of the headache, weakness and numbness had developed in the right hand and had spread to her arm. Neurologic examination was notable for diminished strength and sensation to light touch in the right arm. Owing to concern about acute stroke, magnetic resonance imaging (MRI) and angiography of the brain were performed. The MRI sequence measuring cerebral blood flow indicated hypoperfusion of the left hemisphere that did not match vascular territories (Panel A). Diffusion-weighted imaging showed no ischemic changes (Panel B), and angiography showed patent vessels. A diagnosis of hemiplegic migraine was made. Hemiplegic migraine is an uncommon type of migraine with aura that manifests with reversible motor weakness, along with reversible sensory, visual, or speech changes. The proposed mechanism is cortical spreading depression that results in transient hypoperfusion followed by hyperperfusion in one cerebral cortex. Hemiplegic migraine is usually familial. In this case, family history was negative, so no genetic testing was done. With supportive treatment, the headache resolved in 45 minutes and the arm symptoms in 90 minutes. The patient was discharged with a plan for outpatient follow-up.

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